

	SEEA-Agriculture, Forestry and Fisheries / SEEA-Experimental Ecosystem Accounting
	SEEA-Water / SEEA-physical supply and use tables for water/ SEEA-land accounts / SEEA- ecosystem accounts
	SEEA-Energy
	Material flow accounts / SEEA and tourism satellite accounts
	SEEA-emission accounts
	SEEA-land accounts / SEEA-environmental protection expenditure accounts / SEEA- emission accounts
	SEEA-material flow accounts, water accounts, energy accounts and other resource specific accounts / SEEA-solid waste accounts / SEEA and tourism satellite accounts
	SEEA-Agriculture, Forestry and Fisheries / SEEA-emissions accounts / SEEA-accounts for aquatic resources / SEEA-land accounts / SEEA-environmental taxes and subsidies accounts
	SEEA-land accounts / SEEA-Agriculture, Forestry and Fisheries / ecosystem accounts (ecosystem condition accounts, ecosystem service accounts and biodiversity accounts)

In addition, SDG 3 on Good Health and Wellbeing also shows linkage with SEEA, as all SEEA accounts promote good health and wellbeing. SDG-13 on Climate Change also shows linkage with SEEA and SDG.

45. What is Physical and Monetary Supply and Use Tables?

An economy uses natural resources and other inputs from the environment and lets the environment absorb the by-products. Measuring these flows using physical units can provide instructive information. The physical flow accounting framework provides a set of accounting principles and boundaries within which a consistent recording of all types of physical flows related to economic activity can be made.

The Physical Supply and Use tables are accounts in physical units in the form of matrices that record the flows of natural resources, residuals, products and ecosystems inputs according to origins (supply) and destinations (uses). While the rows of the PSUT show types of natural inputs, products and residuals, the columns reflect both the activity underlying the flow and the economic units involved.

The Physical Supply tables provides information about the flows relating to the production, generation, and supply of natural inputs, products and residuals by different economic units or the environment.

The physical Use tables provides information about the consumption and use of natural inputs, products and residuals by different economic units or the environment.

The Monetary SUTs captures the supply and use of ecosystem services in monetary terms. The intent of Physical flow accounting is to record the physical flows underpinning the monetary supply and use tables and to extend the monetary table to record all physical flows from the environment to the economy and the environment.

46. What are Protected Areas?

A Protected Area²³ is a geographical space, recognized, dedicated and managed through legal or other effective means, to achieve the long-term conservation of nature with associated ecosystem services and cultural values. The establishment of comprehensive, ecologically representative, effectively managed and financially secured protected area networks is a critical strategy not only for biodiversity conservation, but for securing ecosystem goods and services, enabling climate change adaptation and mitigation, and helping countries achieve the Sustainable Development Goals.

Protected areas²⁴ are those in which human occupation or at least the exploitation of resources is limited. The definition that has been widely accepted across regional and global frameworks has been provided by the International Union for Conservation of Nature (IUCN) in its categorization guidelines for protected areas. There are several kinds of protected areas, which vary by level of protection depending on the enabling laws of each country or the regulations of the international organizations involved. The term "protected area" also includes Marine Protected Areas, the boundaries of which will include some area of ocean, and Transboundary Protected Areas that overlap multiple countries which remove the borders inside the area for conservation and economic purposes.

As per IUCN, there are 6 categories of Protected Area; Category Ia & Ib- Strict Nature Reserve and Wilderness Area, Category II- National Park, Category III- Natural Monument or Feature, Category IV- Habitat/Species Managed Area, Category V- Protected Landscape/ Seascape, Category VI-Protected area with sustainable use of natural resources. In India, Protected Areas are declared under Wildlife (Protection) Act 1972 in four categories- National Parks, Wildlife Sanctuaries, Conservation Reserves and Community Reserves.

²³ <https://www.cbd.int/undb/media/factsheets/undb-factsheet-pa-en.pdf>

²⁴ http://www.wiienvi.nic.in/Database/Protected_Area_854.aspx

47. What is Convention on Biological Diversity?

The Convention on Biological Diversity (CBD) is a multilateral agreement of treaty. It was opened for signature on 5th June 1992 during the Earth Summit in Rio de Janeiro, Brazil, and entered into force on 29th December 1993. The prime goals of this convention include sustainability in conservational measures and benefit-sharing along with the development of the strategic plan for the conservational practices.

There are three main objectives of Convention on Biological Diversity (CBD).

- (i) The conservation of biological diversity
- (ii) The sustainable use of the components of biological diversity
- (iii) The fair and equitable sharing of the benefits arising out of the utilization of genetic resources

48. What is CITES and its Appendix?

The Convention on International Trade in Endangered Species of Wild Fauna and Flora (CITES) is a global agreement or multilateral treaty among governments to regulate or ban international trade in species under threat to protect endangered plants and animals. It was adopted by International Union for Conservation of Nature (IUCN) in 1963 and signed in 1973 whereas it was active since 1st July 1975. CITES represents three appendices as Appendices I, II, and III to protect the floral and faunal communities against over-exploitation²⁵.

Appendix I included animals and plants which are threatened with extinction. CITES prohibits international trade in specimens of these species except in exceptional circumstances like when the purpose of trade is not commercial and for scientific research through import and export permit.

Appendix II lists species that are not necessarily now threatened with extinction but that may become so unless trade is closely controlled. The species whose specimens in trade look like those of species listed for conservation reasons. International trade in specimens of Appendix-II species may be authorized by the granting of an export permit or re-export certificate.

Appendix III contains species that are protected in at least one country, which has asked other CITES Parties for assistance in controlling the trade. International trade in specimens of species listed in this Appendix is allowed only on presentation of the appropriate permits or certificates.

²⁵ <https://cites.org/eng/disc/what.php>

49. What are some priority areas identified by NSO, India for Environment Accounting?

NSO, India released “Strategy for Environmental Economic Accounts in India: 2022-2026” in December, 2021. The document has been developed with an aim to implement concerted efforts towards the goal of strengthening the environmental statistics and environmental accounts for India. The strategy document provides a road-map for development of environmental accounting and also helps provide some goals for the short term and also act as a means for measuring India's progress in the domain of the environment accounting. Following priority area has been highlighted in the strategy documents:

- i. Energy Accounts
- ii. Material Flow Accounts
- iii. Ocean Accounts
- iv. Urban Accounts and Biodiversity

50. What are Energy Accounts?

Energy Accounts are a structured compilation of information on flows of energy and its relation with economy and the environment. It is based on SEEA-Energy framework²⁶, which is an accounting approach that records, as completely as possible, the stocks and flows of energy in the territory of reference. It supports analyses of the role of energy within the economy and of the relationship between energy-related activities and the environment. The energy accounts as described in SEEA comprises of primarily three types of accounts:

- (i) Physical supply and use tables: The physical supply and use tables (PSUT) provide a structure for compiling and presenting all energy flows that enter, are used within and leave the national economy of a given country for a period of time.
- (ii) Monetary supply and use tables: Monetary supply and use tables in SEEA fully articulate in monetary terms the flows of energy products in an economy between different economic units.
- (iii) Asset accounts: The purpose of asset accounts is to record the opening and closing stock of known assets and the various types of stock changes over an accounting period.

51. What are Material Flow Accounts?

Economy-wide material flow accounts (EW-MFA) are a statistical accounting framework recording, in thousand tonnes per year, material flows into and out of

²⁶ <https://seea.un.org/seea-energy>

an economy²⁷. They cover solid, gaseous, and liquid materials, except for bulk flows of water and air. The general purpose of EWMFA is to describe the physical interaction of the national economy with the natural environment and the rest of the world economy in terms of flows of materials.

EW-MFA is a statistical framework conceptually embedded in environmental-economic accounts and fully compatible with concepts, principles, and classifications of national accounts – thus enabling a wide range of integrated analyses of environmental, energy and economic issues e.g. through environmental-economic modelling. Material flow accounts are one of the many types of accounts covered by the SEEA Central Framework

52. What are Solid Waste Accounts?

Solid wastes are the discarded material that are no longer required by the owner or user²⁸. These include Chemical and Health care wastes, Radioactive wastes, Metallic wastes, Other recyclables, discarded equipment and vehicles, animal and vegetal wastes, mixed residential and commercial wastes, mineral wastes and soil, combustion wastes and other wastes.

Solid waste accounts, part of SEEA- CF, are organized information on the generation of solid waste and the management of flows of solid waste to recycling facilities, to controlled landfills or directly to the environment. Measures of the amount of waste in aggregate or of quantities of specific waste materials is an important indicator of environmental pressure. The construction of solid waste accounts allows these indicators to be placed in a broader context with economic data in both physical and monetary terms.

53. What are Ocean Accounts?

An Ocean Account is a structured compilation of consistent and comparable information – maps, data, statistics and indicators – concerning marine and coastal environments, including related social circumstances and economic activity²⁹.

The general purpose of such accounts is to inform and enable public policy decision-making about oceans, and related analysis and research. The function of these accounts is to provide coherent structures for standardizing often-fragmented data to produce reliable integrated indicators of interest to policy.

²⁷ <https://seea.un.org/content/material-flow-accounts#:~:text=What%20is%20it%3F,in%20the%20SEEA%20Central%20Framework.>

²⁸ SEEA Central Framework - <https://seea.un.org/content/seea-central-framework>

²⁹ https://unstats.un.org/unsd/statcom/51st-session/documents/BG-item-3h-TG_Ocean%20accounting_ESCAP-E.pdf

Ocean Accounting can (a) help in understanding the role of oceans and coasts in the economy, (b) serve as a mechanism to monitor the investment and net returns from ocean activities, (c) serve as a tool to understand how external events such as storms, climate change and environmental changes can impact Blue Economy development, and (d) help in identification of new areas for investment and innovation.

Ocean accounts can answer the following questions:

- What is the status of ocean wealth including produced assets (e.g. ports) and non-produced assets (e.g. coral reefs, mangroves, fish stocks)? What are the implications of external stresses (e.g., climate change) on ocean wealth?
- How are different groups of people benefitting from the ocean, ocean economy? (income or welfare)

54. What are Urban Accounts?

Ecosystem services are the direct³⁰ and indirect contributions of ecosystems to human well-being. Cities depend on ecosystems both within and beyond the urban environment for a wide variety of goods and services that are essential for economic, social, and environmental sustainability. Ecosystems have the potential in cities to regulate climate, protect against hazards, meet energy needs, support agriculture, prevent soil erosion, and offer opportunities for recreation and cultural inspiration. In many urban areas, particularly in brownfields and other non-used urban land, there are ample opportunities to create novel functioning ecosystems that generate services that enhance the well-being of urban inhabitants.

Ecosystems in urban areas also serve as habitats for species and as storehouses for genetic diversity. Nutrient cycling and soil formation processes are often driven by non-iconic species, such as bacteria or invertebrates; the contribution of biodiversity to these vital ecosystem services often goes unacknowledged or unprotected.

Urban ecosystem accounting provides a framework for quantifying the extent and condition of urban ecosystems and the services they provide and associating these services with beneficiaries.

55. What are the different ecosystem accounts that has been attempted by NSO, India?

NSO, India regularly compiles environment accounts since 2018 following the SEEA framework and has carried out several accounts since then. India has

³⁰ <https://www.cbd.int/doc/health/cbo-action-policy-en.pdf>

developed several extent and condition as well as ecosystem services accounts for various selected ecosystems. A brief description is given below:

Type of account	Topics covered
Ecosystem extent	Change matrix of Land Use – Land Cover (LULC) from 2005-06 to 2011-12 and 2011-12 to 2015-16; and the corresponding Asset Accounts. Land Degradation Account, 2005-06 and 2015-16 Wetland Extent Account, 2006-07 Asset accounts for Minerals and Forests
Ecosystem condition	Soil nutrient indices; Water quality accounts Forest condition account; Cropland condition account
Ecosystem services	Crop provisioning services (monetary) Timber and Non-Timber Forest Products (NTFP) provisioning services (monetary) Carbon retention services provided by forests (physical and monetary) Nature-based tourism (monetary) Soil erosion prevention services provided by croplands (physical)
Thematic Accounts	Biodiversity - Extent of protected areas; State-wise floral and faunal species accounts; Species Richness of IUCN Red List species
Individual environmental asset accounts (SEEA CF)	Forests – Growing Stocks of Timber and Carbon Water Mineral

Apart from this, NSO India constantly strives towards improving the scope and coverage of the Environmental Accounts in India.

56. What are the other organisations involved in Environment accounting in India?

Apart from MoSPI, the natural resource accounting is being dealt by Government Accountings Standards Advisory Board (GASAB) under the aegis of Comptroller and Auditor General of India. In line with its mandate in suggesting accounting framework for enhancing the quality of decision making and public accountability in mind, GASAB has initiated the efforts by preparing a roadmap for implementation of NRA in India. GASAB³¹ has initially proposed preparation of the Asset accounts in the States and then gradually expand it to the nation.

57. To what extent India collaborates with the International agencies regarding environment accounting?

³¹ <http://gasab.gov.in/gasab/pdf/NR-Accounting.pdf>

NSO, India joined the EU funded Natural Capital Accounting and Valuation of Ecosystem Services (NCAVES) India project in 2017 which was jointly implemented by the United Nations Statistics Division (UNSD), United Nations Environment Programme (UNEP) and the Secretariat of the Convention of Biological Diversity (CBD) in five piloted countries-the other countries being Brazil, China, South Africa and Mexico. A large amount of the work on the Environment Accounting in India has been done under the NCAVES project. The report of the project can be accessed at <https://mospi.gov.in/web/mospi/ncaves-india-forum-2021>.

The UN Committee of Experts on Environmental-Economic Accounting (UNCEEA), established by the UN Statistical Commission at its 36th session in March 2005, functions as an intergovernmental body to provide overall vision, coordination, prioritization and direction in the field of environmental economic accounting and supporting statistics. The Bureau of the UNCEEA was established in 2007 to assist the Committee to carry out specific activities pertaining to Environment Accounts. India is a member of the Bureau of UN Committee of Experts on Environmental-Economic Accounting (UNCEEA).

The Global Assessment of Environmental-Economic Accounting and Supporting Statistics is a survey administered under the auspices of the UN Committee of Experts on Environmental-Economic Accounting (UNCEEA). The aim of the Global Assessment is to assess the progress of implementation of the SEEA in countries. NSO, India provides regular responses to the Global Assessment of Environmental-Economic Accounting and Supporting Statistics, conducted UNSD.

In addition, India regularly contributes to SEEA newsletter highlighting the achievements in Environment Statistics and Environment Accounts. NSO, India also actively participates in meetings of London Group on Environment Accounting which serves as a forum to advance methodological research and to develop implementation advice for the SEEA. In addition, NSO India provides critical inputs and feedback on various issues relating to Environment Statistics and Accounts in the annual sessions of UNSC.

58. Where can one find the glossary of all the items used in the EnviStats India publication?

NSO, India has recently released EnviStats India Glossary which encompasses all the terms and definitions used in the EnviStats India publication at a single platform. This will help the user to have a better understanding of the various definition and technical terms used in EnviStats India publications. The glossary will be updated in line with the subsequent release of the publications. The publications can be accessed at

<https://mospi.gov.in/web/mospi/reports-publications/-/reports/view/templateFive/27501?q=RPCAT>



**Social Statistics Division
National Statistical Office
5th Floor, Khurshid Lal Bhawan
Janpath, New Delhi-110001
www.mospi.gov.in**

**For any inputs or suggestions, please contact:
ssd-mospi@gov.in**

